CENG212-Devre Teorisi ve Uygulamaları
H1CD1-Giriş

Dr. Meriç Çetin

versiyon090226

Dersin Kodu	: **CENG212 (3+2) Zorunlu**
Dersin Adı	: Devre Teorisi ve Uygulamaları
Dersi Veren	: Doç. Dr. MERİÇ ÇETİN
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Eposta	: mcetin@pau.edu.tr
Ders saati	: Çarşamba 14.25-16.55 (Teorik), Perşembe 08.55 – 10.35 (Uygulama,Soru Çözümü)
Değerlendirme	: **Vize : %50** **Final: %50**

Dersin Amacı

- Bu dersin amacı temel elektrik ve elektronik devre elemanlarını, bu elemanların karakteristiklerini tanıtmak, devre teoremlerini öğretmek ve bu teoremleri kullanarak öğrencilere bir devreyi analiz etme yeteneğini kazandırmaktır.

Dersin İçeriği

- Elektrik devre değişkenleri
- Devre elemanları, aktif-pasif devre elemanları, bağımlı-bağımsız kaynaklar
- Dirençli devreler, Kirşof kanunları, akım ve gerilim bölücü devreler, devre analizi
- Dirençli devrelerin analiz metotları
- Devre Teoremleri
  - Düğüm gerilimi analizi,
  - Çevre akımları analizi
  - Kaynak dönüşümü,
  - Süperpozisyon,
  - Thevenin/Norton teoremleri,
  - Maksimum güç transferi

Dersin İçeriği

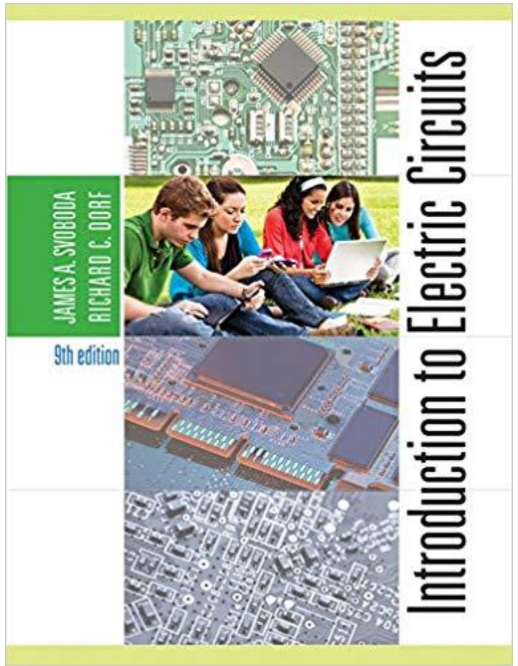
- Enerji depolayan elemanlar: Kapasitörler, bobinler
- Elektroniğe giriş, yükselticiler, yükseltici devre modelleri, işlemsel yükselteçler-Op Amp, ideal Op Amp içeren devre analizleri
- Diyotlar, diyot Devrelerinin analizi, zener diyotlar, doğrultucu devreler, kırpıcı ve bağlayıcı devreler
- İki kutuplu jonksiyon transistörleri-BJTs, aktif moddaki npn ve pnp transistör işlemleri
- DC/AC ortamında transistörlü devre çözümleri
- Alan etkili transistörler-FETs
- DC ortamda alan etkili transistör devre çözümleri

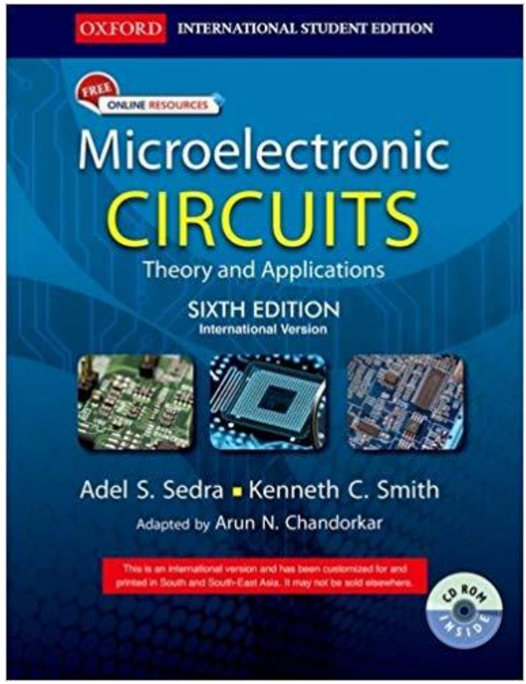
Ön Şartlar

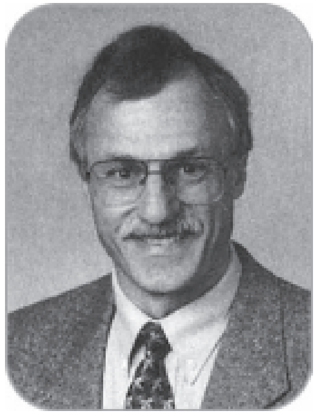
- **Matematik temeli**
  - Diferansiyel Denklemler, Türev, İntegral
- **Doğrusal cebir**
  - Matrisler, Matris İşlemleri
  - Determinant, Doğrusal Denklem Sistemleri
  - Özdeğer, Özvektör
- **İşaret işleme**
- **Beyin-kas koordinasyonu**
- **Analiz ve muhakeme yeteneği**

Kaynaklar

- **Teori için**
  - “**Introduction to Electric Circuits**”, James Svoboda, Richard Dorf
  - “**Microelectronic Circuits**”, Adel S. Sedra, Kenneth C. Smith
- **Uygulamaya Meraklı Öğrenciler için**
  - **“Arduino Robotics”,** John D. Warren, Josh Adams, Harald Molle
  - **“Beginning Arduino”,** Michael McRoberts
  - **“30 Arduino Projects for the Evil Genius”**, Simon Monk
- **EDS duyurularını takip ediniz.**







James A. Svoboda is an associate professor of electrical and computer engineering at Clarkson University, where he teaches courses on topics such as circuits, electronics, and computer programming. He earned a PhD in electrical engineering from the University of Wisconsin at Madison, an MS from the University of Colorado, and a BS from General Motors Institute.

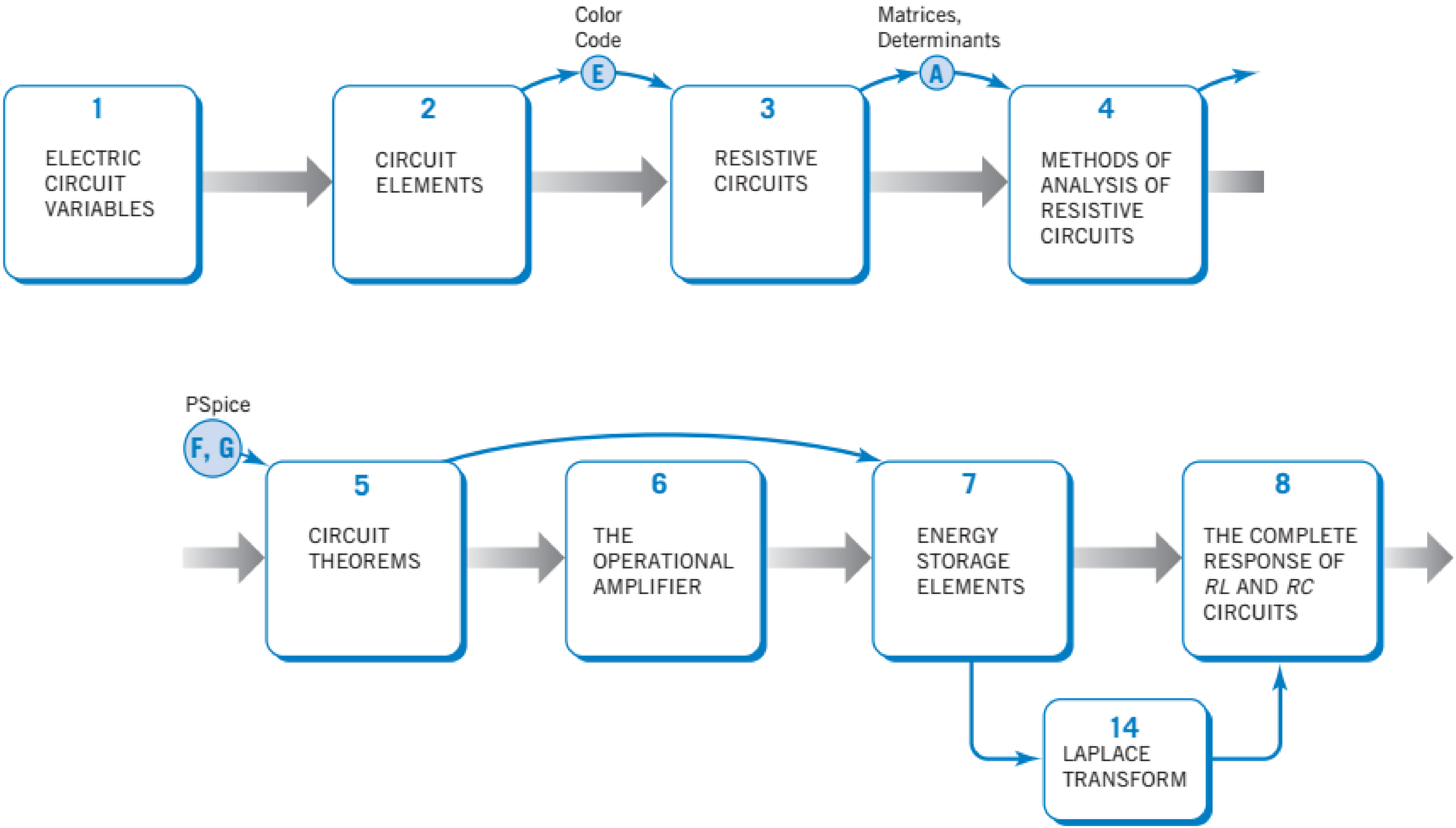
Sophomore Circuits is one of Professor Svoboda's favorite courses. He has taught this course to 6,500 undergraduates at Clarkson University over the past 35 years. In 1986, he received Clarkson University's Distinguished Teaching Award.

Professor Svoboda has written several research papers describing the advantages of using nullors to model electric circuits for computer analysis. He is interested in the way technology affects engineering education and has developed several software packages for use in Sophomore Circuits.

Richard C. Dorf, professor of electrical and computer engineering at the University of California, Davis, teaches graduate and undergraduate courses in electrical engineering in the fields of circuits and control systems. He earned a PhD in electrical engineering from the U.S. Naval Postgraduate School, an MS from the University of Colorado, and a BS from Clarkson University. Highly concerned with the discipline of electrical engineering and its wide value to social and economic needs, he has written and lectured internationally on the contributions and advances in electrical engineering.

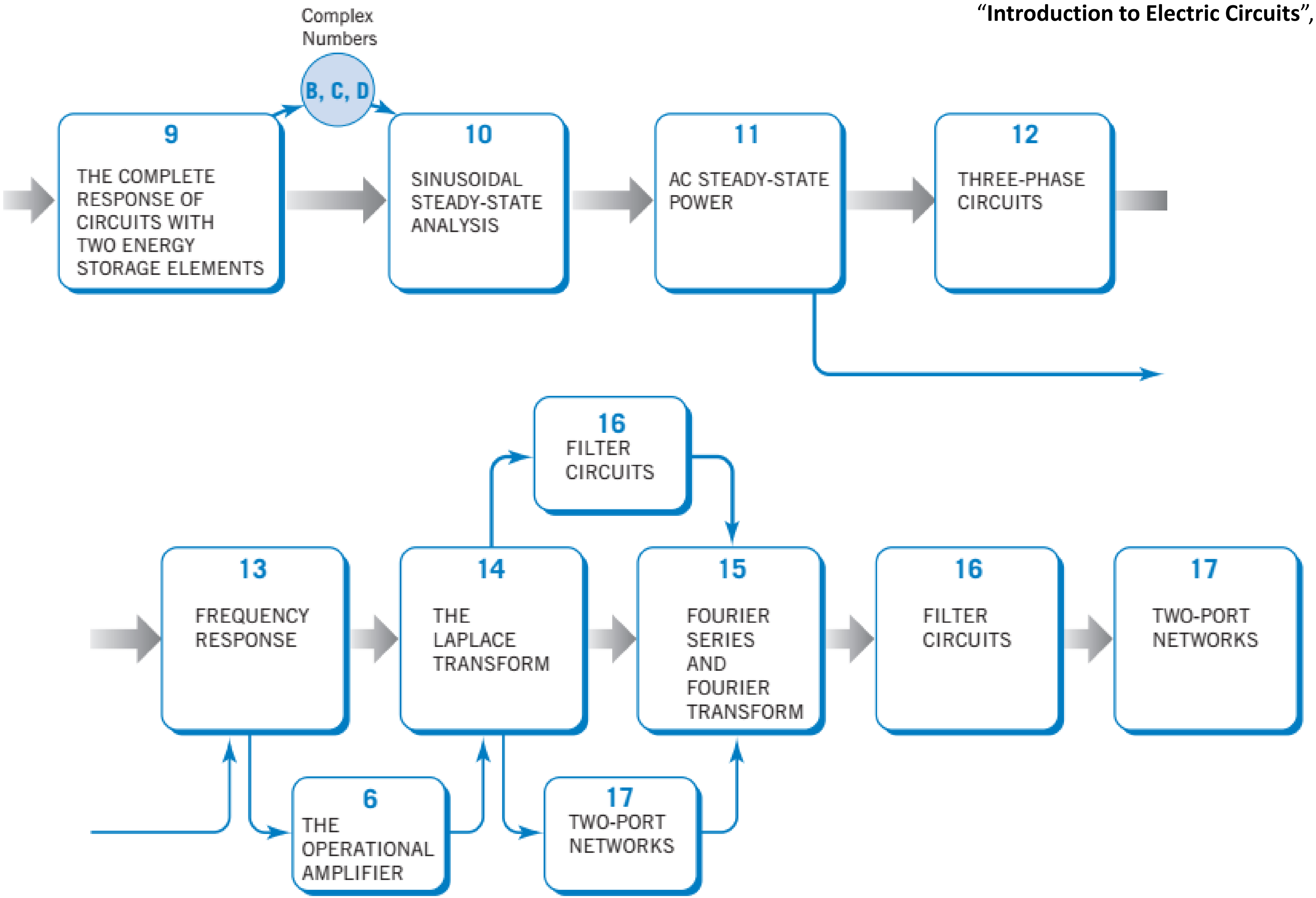
Professor Dorf has extensive experience with education and industry and is professionally active in the fields of robotics, automation, electric circuits, and communications. He has served as a visiting professor at the University of Edinburgh, Scotland, the Massachusetts Institute of Technology, Stanford University, and the University of California at Berkeley.

A Fellow of the Institute of Electrical and Electronic Engineers and the American Society for Engineering Education, Dr. Dorf is widely known to the profession for his *Modern Control Systems*, twelfth edition (Pearson, 2011) and *The International Encyclopedia of Robotics* (Wiley, 1988). Dr. Dorf is also the coauthor of *Circuits, Devices and Systems* (with Ralph Smith), fifth edition (Wiley, 1992). Dr. Dorf edited the widely used *Electrical Engineering Handbook*, third edition (CRC Press and IEEE press), published in 2011. His latest work is Technology Ventures, fourth edition (McGraw-Hill 2013).



“**Introduction to Electric Circuits**”, j., Svoboda & R., Dorf

“**Introduction to Electric Circuits**”, j., Svoboda & R., Dorf



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“**Introduction to Electric Circuits**”, j., Svoboda & R., Dorf

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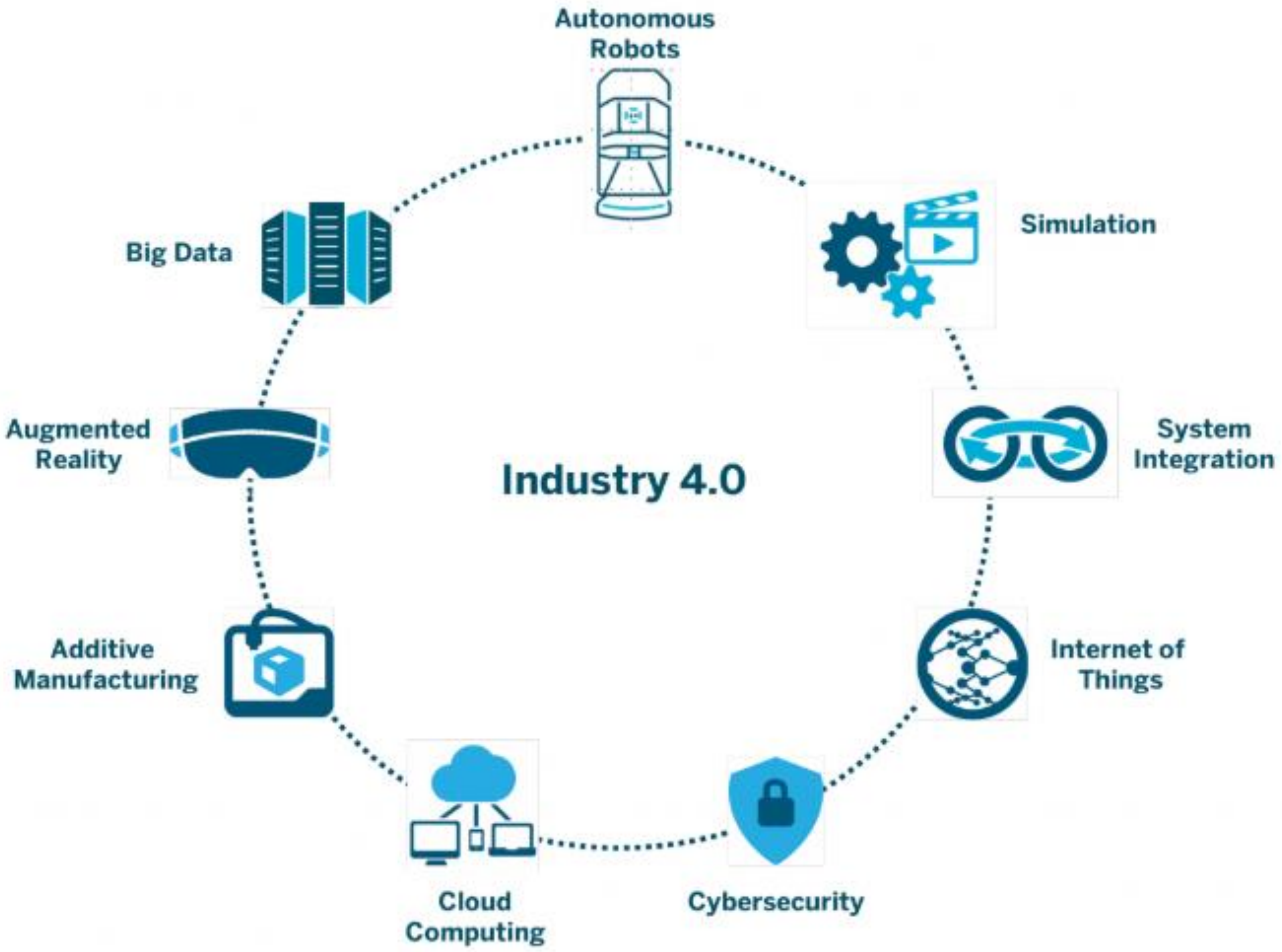
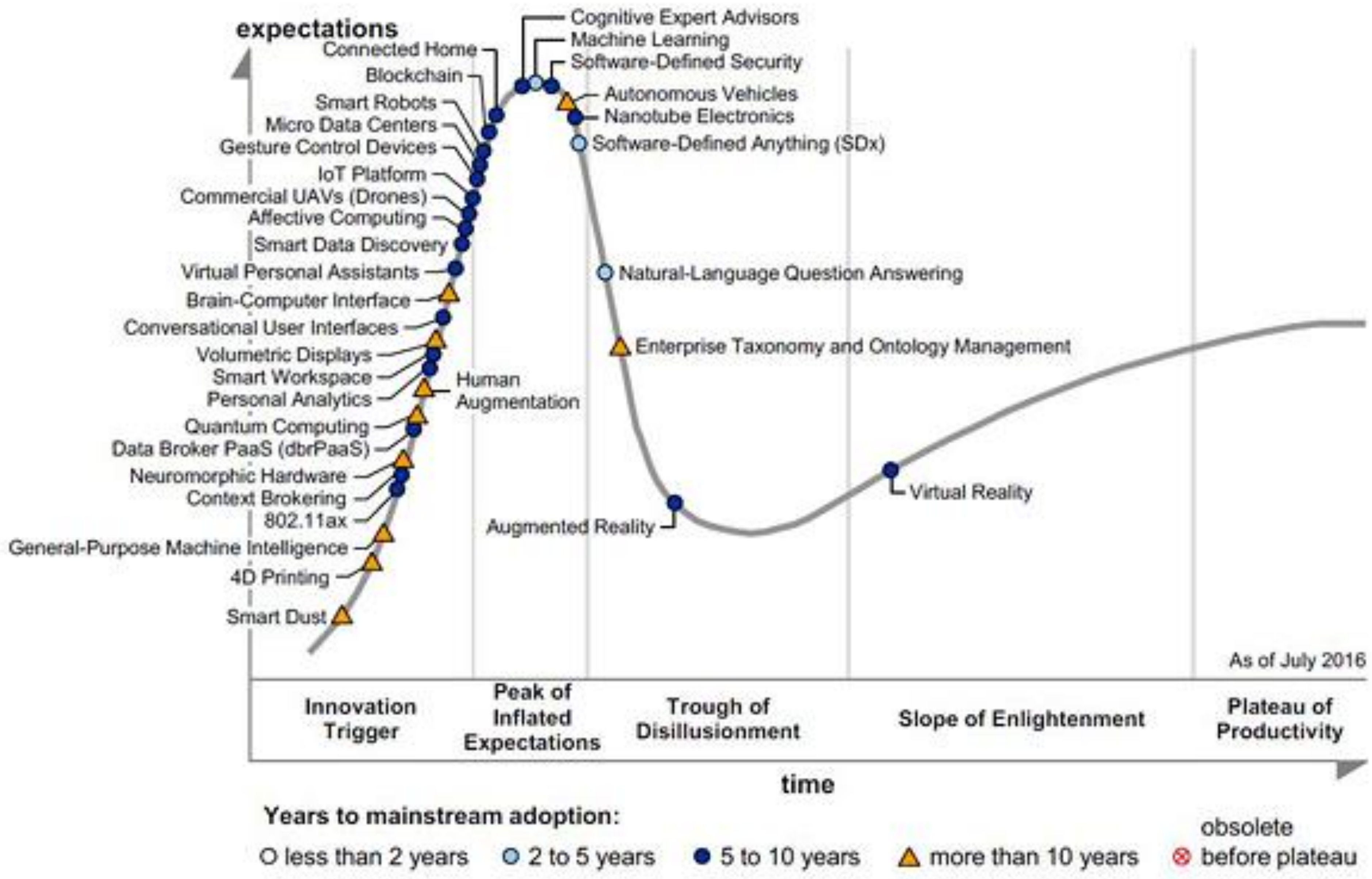


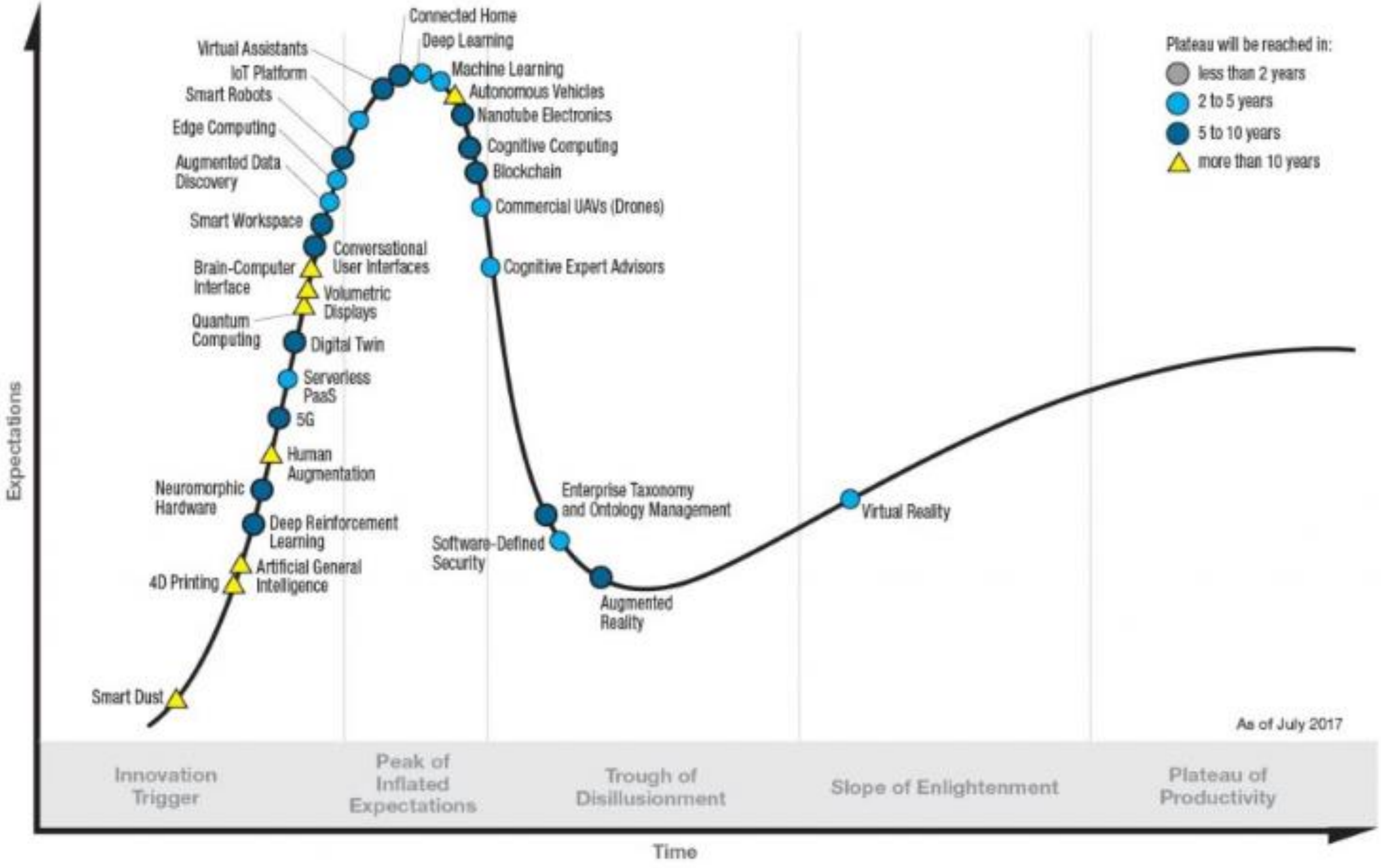
Autonomous Robots
Simulation
System Integration
Internet of Things
Cybersecurity
Cloud Computing
Additive Manufacturing
Augmented Reality
Big Data
Industry 4.0

Hype Cycle for Emerging Technologies, 2016

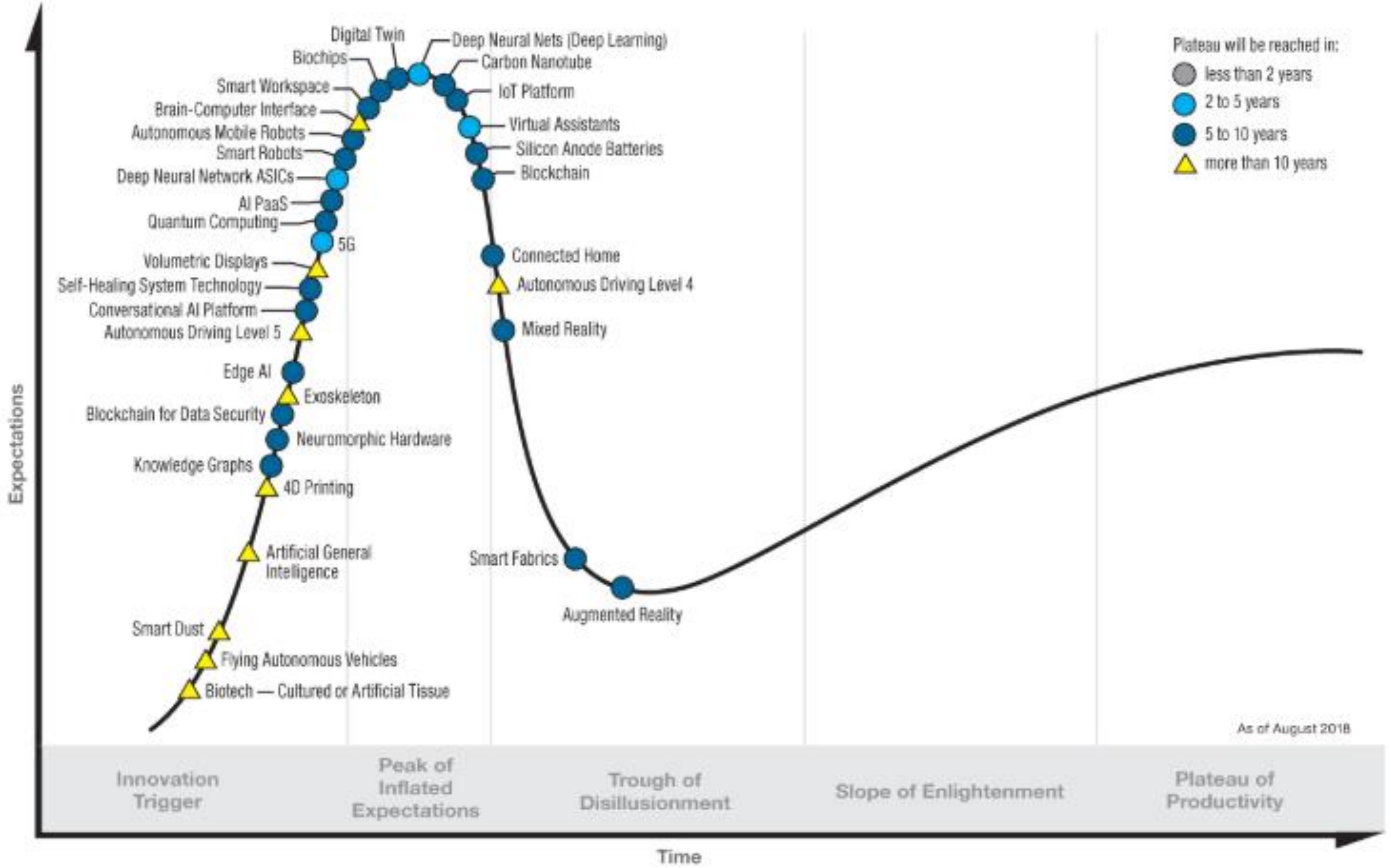


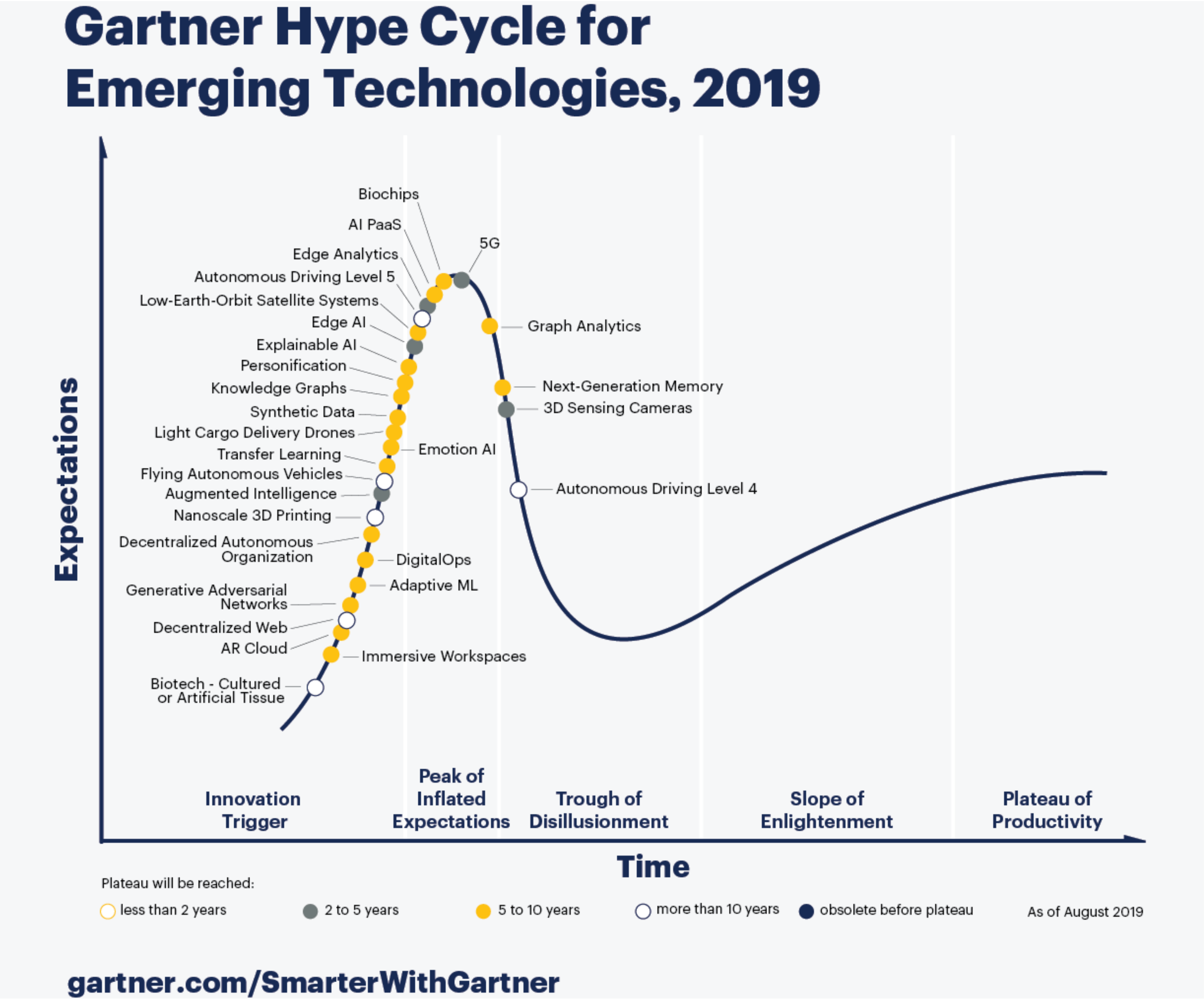
Source: Gartner (July 2016)

Gartner Hype Cycle for Emerging Technologies, 2017

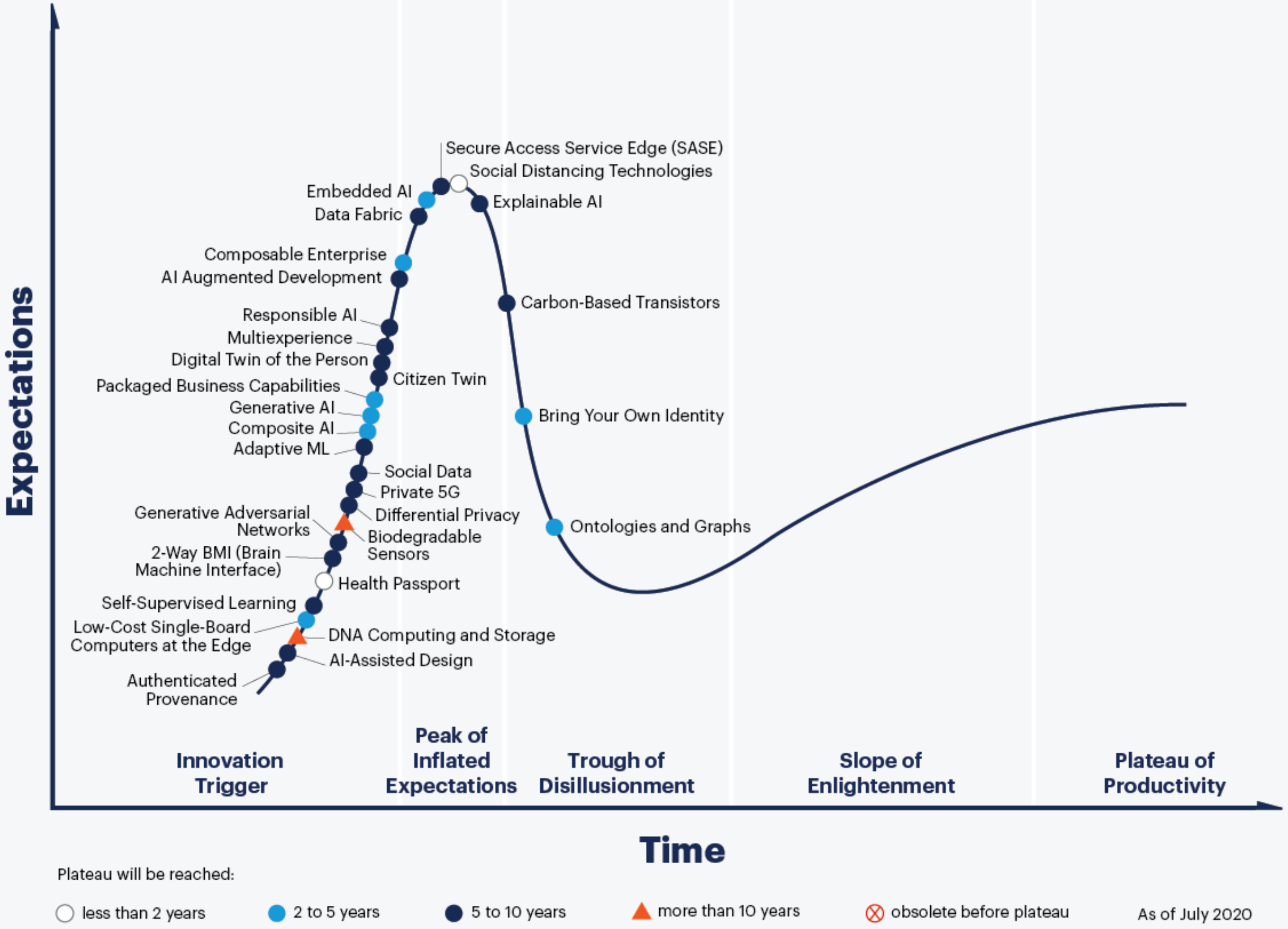


Hype Cycle for Emerging Technologies, 2018




Gartner Hype Cycle for
Emerging Technologies, 2019
Biochips
AI PaaS
Edge Analytics
Autonomous Driving Level 5
Low-Earth-Orbit Satellite Systems
Edge AI
Explainable AI
Personification
Knowledge Graphs
Synthetic Data
Light Cargo Delivery Drones
Transfer Learning
Flying Autonomous Vehicles
Augmented Intelligence
Nanoscale 3D Printing
Decentralized Autonomous Organization
Generative Adversarial Networks
Decentralized Web
AR Cloud
Biotech - Cultured or Artificial Tissue
5G
Graph Analytics
Next-Generation Memory
3D Sensing Cameras
Emotion AI
Autonomous Driving Level 4
DigitalOps
Adaptive ML
Immersive Workspaces
Expectations
Innovation Trigger
Peak of Inflated Expectations
Trough of Disillusionment
Slope of Enlightenment
Plateau of Productivity
Time
Plateau will be reached:
less than 2 years
2 to 5 years
5 to 10 years
more than 10 years
obsolete before plateau
As of August 2019
gartner.com/SmarterWithGartner

Hype Cycle for Emerging Technologies, 2020

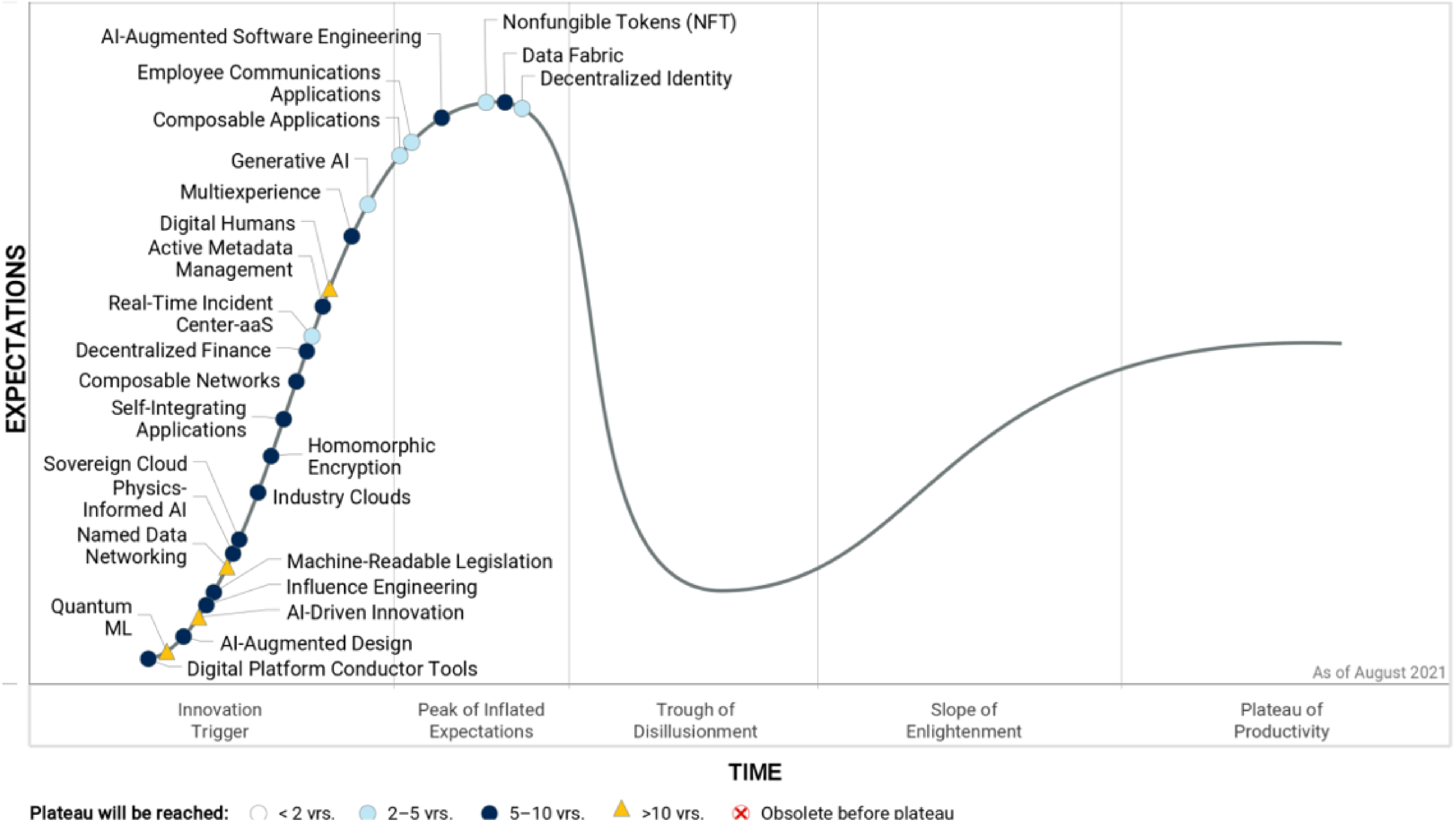


gartner.com/SmarterWithGartner

Source: Gartner

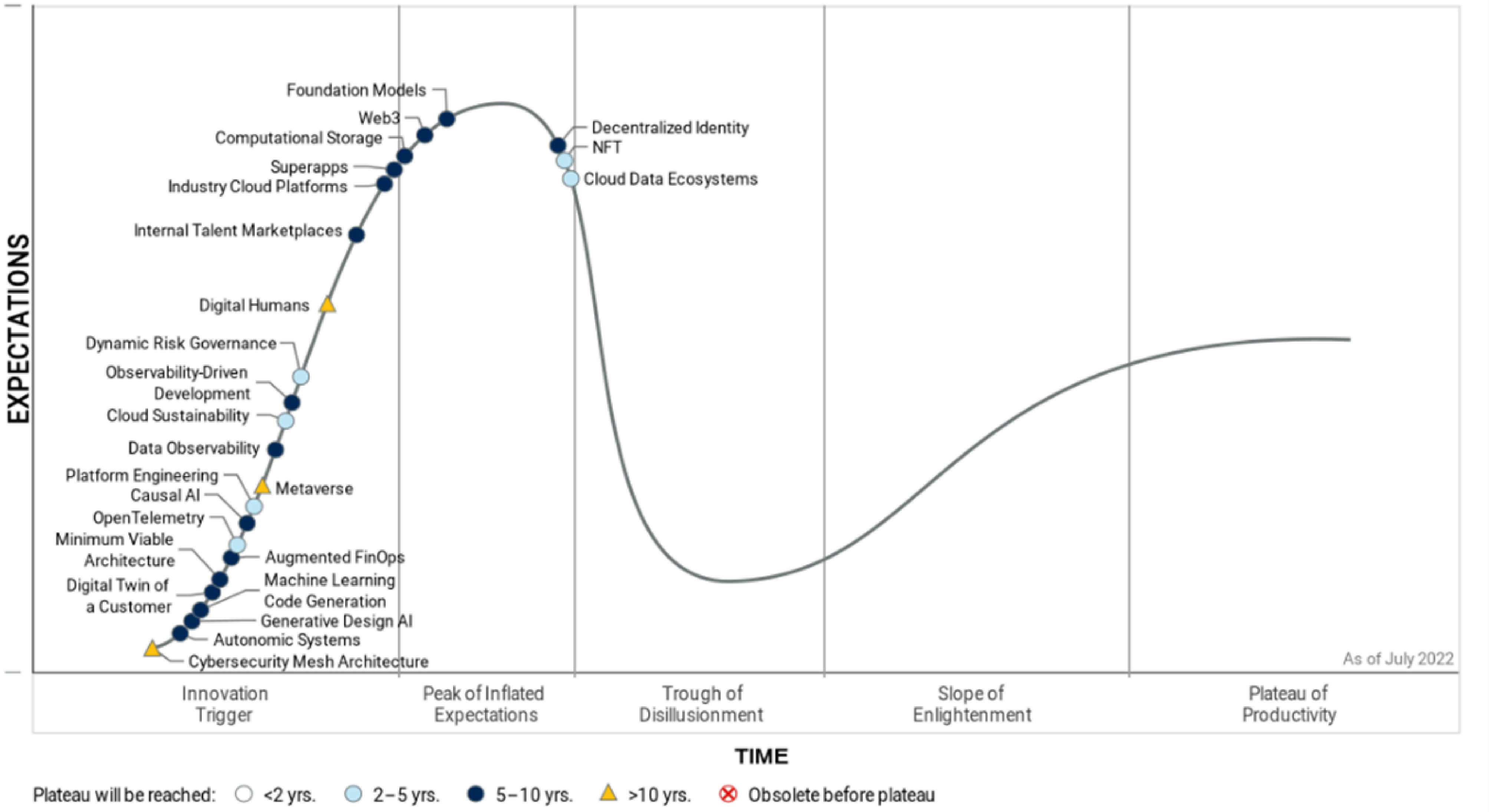
Gartner

Hype Cycle for Emerging Technologies, 2021



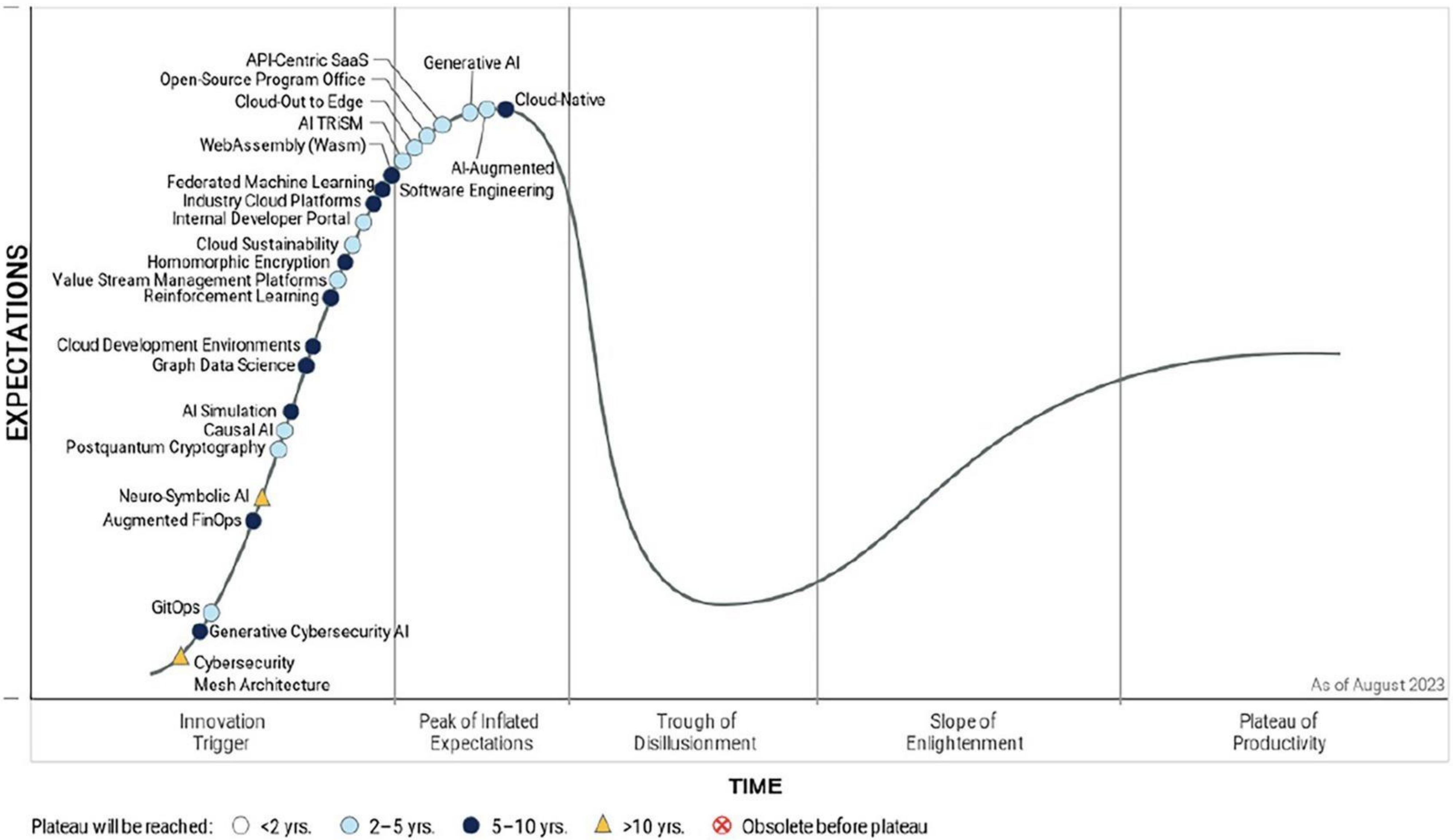
Source: Gartner (August 2021)

Hype Cycle for Emerging Technologies, 2022



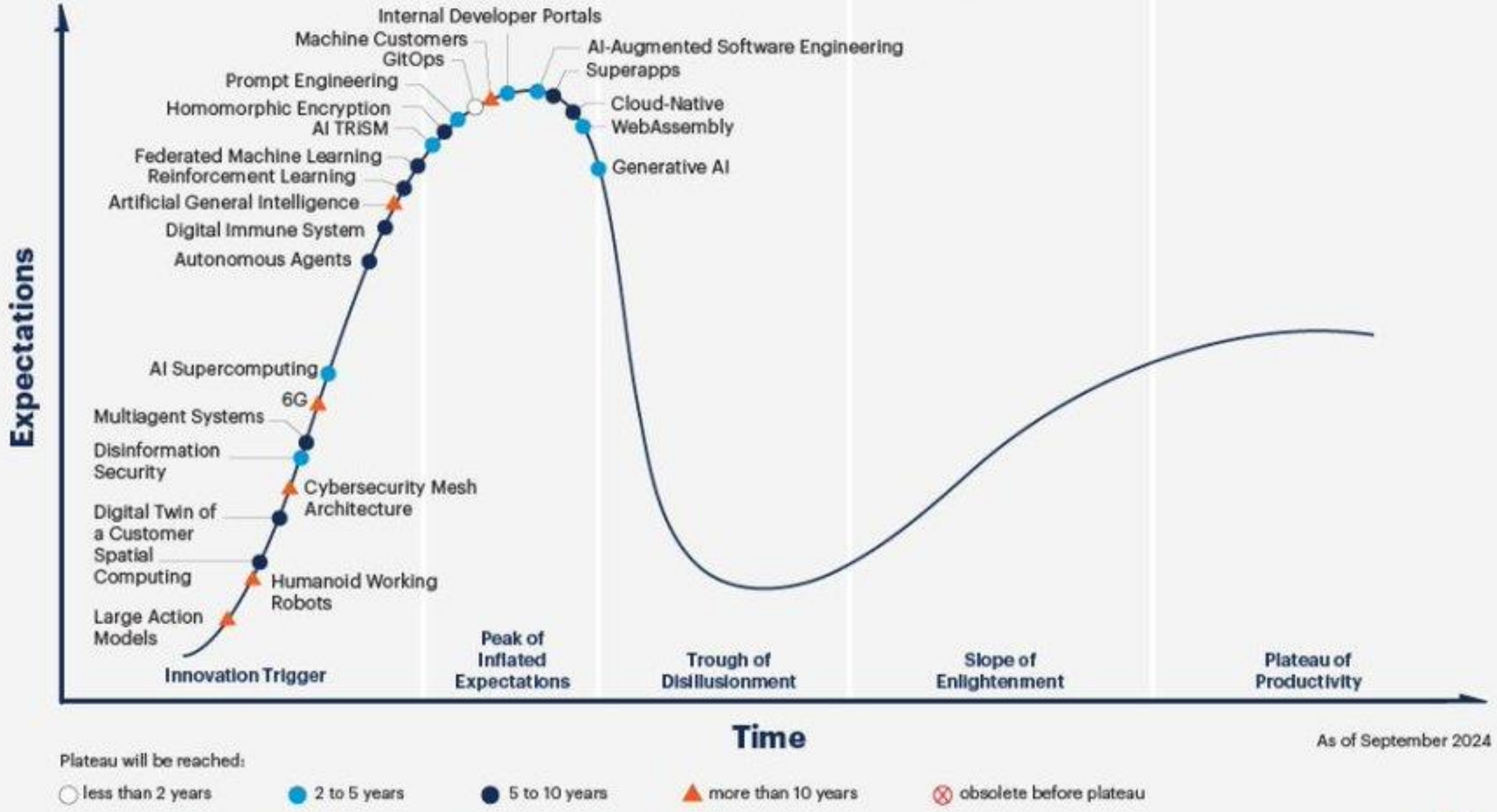
Gartner

Hype Cycle for Emerging Technologies, 2023



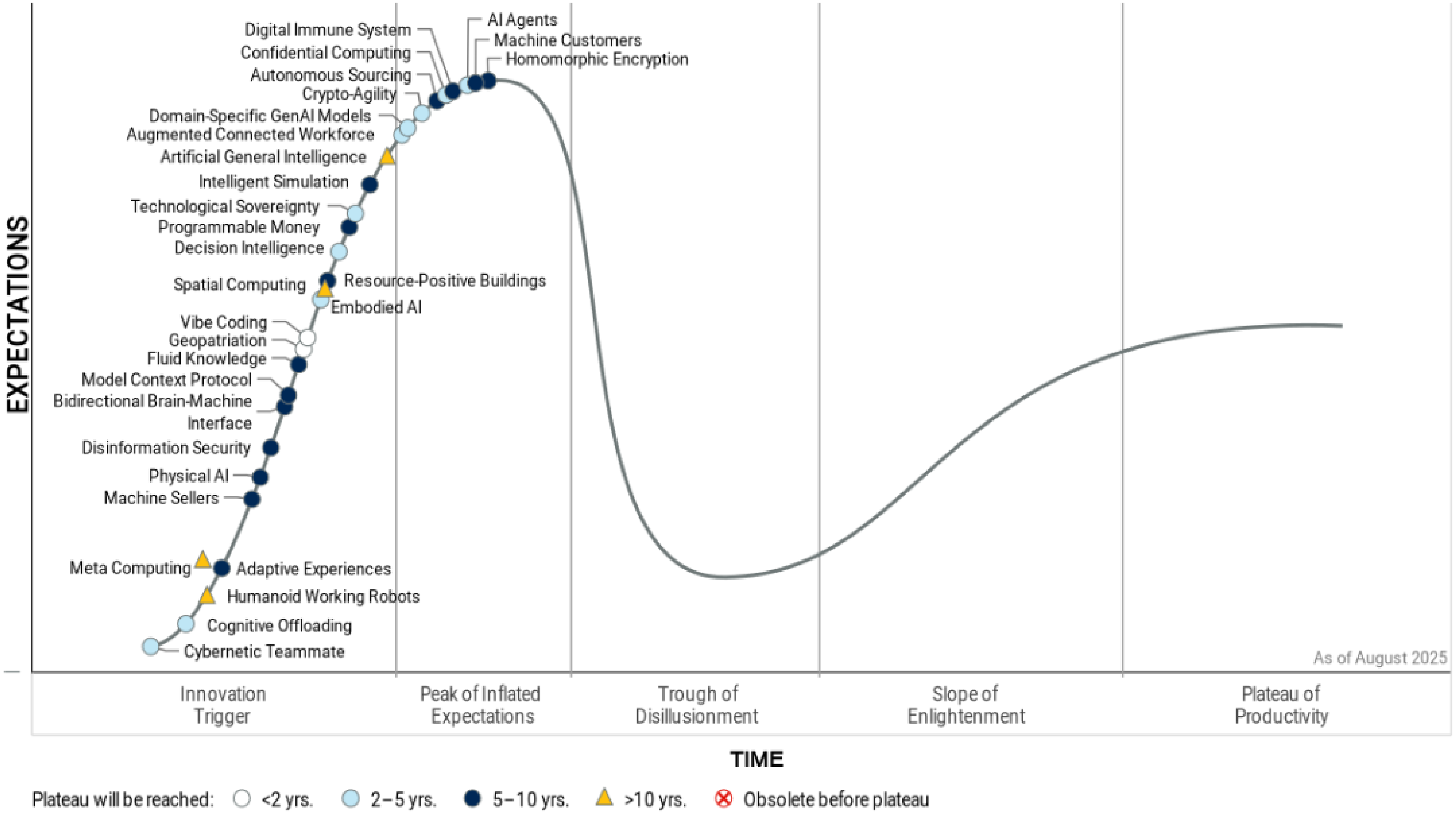
Gartner

Hype Cycle for Emerging Technologies, 2024



Source: Gartner

Hype Cycle for Emerging Technologies 2025



Gartner

Timeline of electrical and electronic engineering

600 B.C.	Thales of Miletus discovered static electricity by rubbing fur on substances such as amber
1600	English scientist William Gilbert coined the word *electricus* after careful experiments.
1705	English scientist Francis Hauksbee made a glass ball that glowed when spun and rubbed with the hand
1720	English scientist Stephen Gray made the distinction between insulators and conductors
1745	German physicist Ewald Georg von Kleist and Dutch scientist Pieter van Musschenbroek invented Leyden jars
1752	American scientist Benjamin Franklin showed that lightning was electrical by flying a kite, and explained how Leyden jars work
1780	Italian scientist Luigi Galvani discovered the Galvanic action in living tissue

Timeline of electrical and electronic engineering

1785	French physicist Charles-Augustin de Coulomb formulated and published Coulomb's law in his paper Premier Mémoire sur l'Électricité et le Magnétisme
1785	French mathematician Pierre-Simon Laplace developed the Laplace transform to transform a linear differential equation to an algebraic equation. Later, his transform became a tool in circuit analysis.
1800	Italian physicist Alessandro Volta invented the battery
1808	Atomic theory by John Dalton
1816	English inventor Francis Ronalds built the first working electric telegraph
1820	Danish physicist Hans Christian Ørsted accidentally discovered that an electric field creates a magnetic field
1820	One week after Ørsted's discovery, French physicist André-Marie Ampère published his law. He also proposed right-hand screw rule

Timeline of electrical and electronic engineering

1821	German scientist Thomas Johann Seebeck discovered thermoelectricity
1825	English physicist William Sturgeon developed the first electromagnet
1827	German physicist Georg Ohm introduced the concept of electrical resistance
1831	English physicist Michael Faraday published the law of induction (Joseph Henry developed the same law independently)
1831	American scientist Joseph Henry in United States developed a prototype DC motor
1832	French instrument maker Hippolyte Pixii in France developed a prototype DC generator
1833	Michael Faraday developed laws of electrolysis
1833	Michael Faraday invented thermistor

Timeline of electrical and electronic engineering

1833	English Samuel Hunter Christie invented Wheatstone bridge (It is named after Charles Wheatstone who popularized it)
1836	Irish priest (and later scientist) Nicholas Callan invented transformer in Ireland
1837	English scientist Edward Davy invented the electric relay
1839	French scientist Edmond Becquerel discovered the Photovoltaic Effect
1844	American inventor Samuel Morse developed telegraphy and the Morse code
1845	German physicist Gustav Kirchhoff developed two laws now known as Kirchhoff's Circuit laws
1850	Belgian engineer Floris Nollet invented (and patented) a practical AC generator

Timeline of electrical and electronic engineering

1851	Heinrich Daniel Ruhmkorff first coil, which he patented in 1851
1855	First utilization of AC (in electrotherapy) by French neurologist Guillaume Duchenne
1856	Belgian engineer Charles Bourseul proposed telephony
1856	First electrically powered light house in England
1860	German scientist Johann Philipp Reis invented Microphone
1862	Scottish physicist James Clerk Maxwell published four equations bearing his name
1866	Transatlantic telegraph cable
1873	Belgian engineer Zenobe Gramme who developed DC generator accidentally discovered that a DC generator also works as a DC motor during an exhibit in Vienna.

Timeline of electrical and electronic engineering

1876	Russian engineer Pavel Yablochkov invented electric carbon arc lamp
1876	Scottish inventor Alexander Graham Bell patented the telephone
1877	First street lighting in Paris, France
1877	American inventor Thomas Alva Edison invented phonograph
1877	German industrialist Werner von Siemens developed primitive loudspeaker
1878	First hydroelectric plant in Cragside, England

Timeline of electrical and electronic engineering

1878	English engineer Joseph Swan invented Incandescent light bulb
1879	American physicist Edwin Herbert Hall discovered Hall Effect
1879	Thomas Alva Edison introduced a long lasting filament for the incandescent lamp.
1880	French physicists Pierre Curie and Jacques Curie discovered Piezoelectricity
1882	First thermal power stations in London and New York
1883	English physicist J J Thomson invented waveguides
1887	German American inventor Emile Berliner invented gramophone record
1888	German physicist Heinrich Hertz proves the existence of electromagnetic waves, including what would come to be called radio waves.

Timeline of electrical and electronic engineering

1888	Italian physicist and electrical engineer Galileo Ferraris publishes a paper on the induction motor and Serbian-American engineer Nikola Tesla gets a US patent on the same device[3][4]
1890	Thomas Alva Edison invents the fuse
1893	During the Fourth International Conference of Electricians in Chicago electrical units were defined
1894	Russian physicist Alexander Stepanovich Popov finds a use for radio waves, building a radio receiver that can detect lightning strikes
1895	Discovery of X-rays by Wilhelm Röntgen
1896	First successful intercontinental telegram
1897	German inventor Karl Ferdinand Braun invented cathode ray oscilloscope (CRO)
1900	Italian inventor Guglielmo Marconi builds first radio communication system, based on radiotelegraphy

Timeline of electrical and electronic engineering

1901	First transatlantic radio transmission by Guglielmo Marconi
1901	American engineer Peter Cooper Hewitt invented Fluorescent lamp
1904	English engineer John Ambrose Fleming invented diode
1906	American inventor Lee de Forest invented triode
1908	Scottish engineer Alan Archibald Campbell-Swinton, laid the principles of Television.
1911	Dutch physicist Heike Kamerlingh Onnes discovered Superconductivity
1912	American engineer Edwin Howard Armstrong developed Electronic oscillator
1915	French physicist Paul Langevin and Russian engineer Constantin Chilowsky invented sonar
1917	American engineer Alexander M. Nicholson invented crystal oscillator

Timeline of electrical and electronic engineering

1918	French physicist Henri Abraham and Eugene Bloch invented multivibrator
1919	Edwin Howard Armstrong developed standard AM radio receiver
1921	Metre Convention was extended to include the electrical units
1921	Edith Clarke invents the "Clarke calculator", a graphical calculator for solving line equations involving hyperbolic function, allowing electrical engineers to simplify calculations for inductance and capacity in power transmission lines[5]
1925	Austrian American engineer Julius Edgar Lilienfeld patented the first FET (which became popular much later)
1926	Yagi-Uda antenna was developed by the Japanese engineers Hidetsugu Yagi and Shintaro Uda
1927	American engineer Harold Stephen Black invented negative feedback amplifier

Timeline of electrical and electronic engineering

1927	German Physicist Max Dieckmann invented Video camera tube
1928	First experimental Television broadcast in the US.
1929	First public TV broadcast in Germany
1931	First wind energy plant in the Soviet Union
1936	Dudley E. Foster and Stuart William Seeley developed FM detector circuit.
1936	Austrian engineer Paul Eisler invented Printed circuit board
1936	Scottish Scientist Robert Watson-Watt developed the Radar concept which was proposed earlier.

Timeline of electrical and electronic engineering

1938	Russian American engineer Vladimir K. Zworykin developed Iconoscope
1939	Edwin Howard Armstrong developed FM radio receiver
1939	Russell and Sigurd Varian developed the first Klystron tube in the US.
1941	German engineer Konrad Zuse developed the first programmable computer in Berlin
1944	Scottish Engineer John Logie Baird developed the first color picture tube
1945	Transatlantic telephone cable
1947	American engineers John Bardeen and Walter Houser Brattain together with their group leader William Shockley invented transistor.

Timeline of electrical and electronic engineering

1948	Hungarian-British physicist Dennis Gabor invented Holography
1950	French physicist Alfred Kastler invented MASER
1951	First nuclear power plant in the US
1953	First fully transistorized computer in the US
1958	American engineer Jack Kilby invented the integrated circuit (IC)
1960	American engineer Theodore Harold Maiman invented the LASER
1962	Nick Holonyak Jr. invented the LED
1963	First home Videocassette recorder (VCR)
1963	Electronic calculator
2008	American scientist Richard Stanley Williams invented memristor which was proposed by Leon O. Chua in 1971